

BLUESTOCK FINTECH

Mutual Fund Performance Analysis

Capstone Project — Final Report

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1. Executive Summary

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This report presents the findings of a comprehensive Mutual Fund (MF) performance analysis conducted during a Data Analyst internship at Bluestock Fintech. The project involved designing and implementing a full end-to-end data pipeline — from raw data ingestion through ETL processing, exploratory analysis, and interactive Power BI dashboard visualisation — to surface actionable insights about mutual fund investment behaviour, fund performance trends, risk profiles, and investor demographics.

The analysis was conducted over 7 days using **10 structured datasets** containing 32,778 transactions from 5,000 investors with a combined investment amount of **■4 Billion INR**. The project culminated in a production-ready **4-page Power BI dashboard**, a Python-based ETL pipeline, and advanced analytics covering VaR, CVaR, Sharpe Ratio, SIP Continuity, Cohort Analysis, and a Fund Recommendation System.

Key Highlights

Metric	Value	Observation
Total Transactions	32,778	Across all transaction types
Total Investment Amount	■4 Billion INR	Lumpsum contributes highest share
Total Investors	5,000	Across multiple states and age groups
Datasets Used	10	Structured CSV files
Dashboard Tool	Power BI	4-page interactive dashboard
Analytics Tool	Python (Pandas, NumPy)	Full ETL + advanced analytics
Top Investing States	Punjab, Tamil Nadu, MP	Highest transaction volume
Highest Transaction Year	2024	Higher volume than 2025
Advanced Analytics	VaR, CVaR, Sharpe, SIP, Cohort, Fund Recommender	All completed
Deliverables Completed	7 / 7	All Day 7 objectives met

2. Data Sources

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The dataset for this project comprised **10 structured CSV files** provided as part of the Bluestock Fintech internship, covering NAV history, investor transactions, fund performance metrics, fund scorecards, AUM data, SIP inflows, category inflows, industry folio counts, portfolio holdings, and benchmark indices.

2.1 Primary Datasets

- 01_fund_master.csv — Fund master reference data
- 02_nav_history.csv — Daily Net Asset Values per fund and date
- 03_aum_by_fund_house.csv — AUM data by fund house
- 04_monthly_sip_inflows.csv — Monthly SIP inflow data
- 05_category_inflows.csv — Category-level inflow data
- 06_industry_folio_count.csv — Industry-level folio counts
- 07_scheme_performance.csv — Fund-level performance metrics (CAGR, Sharpe, drawdown)
- 08_investor_transactions.csv — Investor transaction records including amount, type, and date
- 09_portfolio_holdings.csv — Portfolio holding details per fund
- 10_benchmark_indices.csv — Benchmark index data for performance comparison

2.2 Key Columns Used

- amfi_code — Unique fund identifier
- nav — Net Asset Value (daily)
- transaction_date — Date of investor transaction
- amount_inr — Transaction amount in Indian Rupees
- state — Investor's state of residence
- age_group — Investor age bracket
- investor_id — Unique investor identifier
- transaction_type — SIP, Lumpsum, Redemption, etc.

2.3 Data Volume

Metric	Value
Total Transactions	32,778
Total Investors	5,000
Total Investment Amount	■4 Billion INR
Total Datasets	10 structured CSV files
Date Range	2023 – 2025
Update Frequency	Batch (provided as static CSVs)

3. ETL Design

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The ETL pipeline was built entirely in Python using Pandas, with CSV files as both input and output. The pipeline handles all data quality issues and prepares clean, analysis-ready datasets.

3.1 Tools Used

- Python (Pandas) — Primary ETL and data cleaning tool
- NumPy — Numerical computations
- Matplotlib / Seaborn — EDA visualisation
- Power BI — Dashboard and reporting

3.2 Transformation Steps

Step	Action	Detail
1	Data Cleaning	Loaded all 10 CSVs into Pandas DataFrames
2	Date Conversion	Parsed transaction_date and nav_date to datetime format
3	Duplicate Removal	Dropped duplicate rows using investor_id + transaction_date as composite key
4	Missing Value Handling	Forward-filled NAV series; flagged records with null amount_inr
5	Feature Engineering	Computed transaction year/month, age group bins, investment category flags
6	CSV Exports	Exported cleaned datasets to /data/processed/ for Power BI and analysis

4. EDA Findings

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Exploratory analysis was performed using Python (Pandas, Matplotlib, Seaborn) and cross-validated through Power BI visualisations. The following summarises the key findings derived directly from the dashboard and analytical outputs.

4.1 Key Metrics

Metric	Value
Total Investment Amount	■4 Billion INR
Total Investors	5,000
Total Transactions	32,778
Highest Share of Investment	Lumpsum (highest share; SIP smallest share)
Top Investing States	Punjab, Tamil Nadu, Madhya Pradesh
Year with Higher Transaction Volume	2024 (higher than 2025)

4.2 Transaction Type Analysis

Lumpsum transactions contributed the highest share of investment volume, while SIP contributed the smallest share. This pattern indicates a preference among investors for bulk one-time investments in terms of total value, even as SIP transactions remain the most frequent by count, reflecting the popularity of systematic investment plans.

4.3 Geographic Distribution

Punjab, Tamil Nadu, and Madhya Pradesh emerged as the top three investing states by transaction volume. This geographic concentration highlights regional patterns in mutual fund adoption and may inform targeted investor outreach strategies.

4.4 Year-on-Year Trends

Transaction volume in 2024 was higher than in 2025, suggesting either a peak in investor activity during 2024 or an incomplete data capture for 2025 within the provided dataset. This trend was confirmed across both transaction count and investment amount dimensions.

5. Performance Analysis

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Advanced performance analytics were developed using Python as part of the Day 6 deliverables. These metrics provide a deeper, risk-adjusted view of fund performance beyond simple return figures.

5.1 Metrics Computed

- CAGR Analysis — Compound Annual Growth Rate computed per fund
- Sharpe Ratio — Risk-adjusted return relative to a benchmark rate
- Sortino Ratio — Downside deviation-adjusted return metric
- Maximum Drawdown — Largest peak-to-trough decline per fund
- Risk Classification — Funds categorised as Low / Medium / High risk based on volatility
- Fund Recommendation System — Rule-based recommender using Sharpe, CAGR, and drawdown scores

5.2 Advanced Analytics

Analysis	Description
Historical VaR & CVaR	Value at Risk and Conditional VaR computed at 95% confidence level
Rolling 90-Day Sharpe	Rolling Sharpe Ratio tracked over time; chart generated in Matplotlib
Investor Cohort Analysis	Investors grouped by join cohort; retention and activity tracked
SIP Continuity Analysis	SIP investors tracked for payment continuity and drop-off patterns
HHI Concentration	Herfindahl-Hirschman Index computed to measure fund concentration
Fund Recommendation Engine	Rule-based system recommending funds based on risk profile and score

6. Dashboard Overview

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An interactive **4-page Power BI dashboard** (bluestock_mf.pbix) was developed to enable exploration of investor behaviour, fund performance, and market trends without requiring access to the underlying Python environment.

6.1 Dashboard Pages

Page	Title	Key Visuals
1	Industry Overview	Total Investment KPI, Investor Count, Transaction Distribution, State-wise Investment map
2	Fund Performance	CAGR by Fund, Sharpe Ratio Table, Fund Score Analysis
3	Investor Analytics	State Filter, Age Group Filter, Investment Trend by Year, State-wise Investment
4	SIP & Market Trends	Transaction Type Analysis, Investment Trends, Geographic Analysis

6.2 Key Interactivity Features

- State slicer filters all visuals on Investor Analytics page simultaneously
- Age group filter for demographic drill-down across investment trends
- Transaction type toggle: SIP / Lumpsum / Redemption / Switch
- Year filter for 2023–2025 trend comparison
- Tooltips with investor and fund details on all chart elements

7. Limitations

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While the analysis provides robust insights, several limitations should be acknowledged:

- Data completeness: Amount and investor data had minor gaps for certain records. Forward-fill and flag-based imputation methods were applied.
- Static dataset: The pipeline operates on batch CSV files. There is no real-time or live data feed integrated in this version.
- Time horizon constraint: The dataset covers a limited date range. Longer historical data would strengthen trend analysis.
- No real-time data: NAV data reflects snapshot pricing and is not suitable for intraday or live tracking.
- Geographic granularity: State-level analysis does not extend to city or district level, limiting hyper-local insights.
- Market regime bias: The period covered may include unusual market conditions that skew certain performance metrics.

8. Recommendations

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Based on the analytical findings, the following recommendations are proposed for investors and for future enhancements to this project:

8.1 Investor-Facing Recommendations

- Continue SIP investments for consistency — cohort analysis shows SIP investors demonstrate more stable long-term participation
- Diversify across risk grades — the fund scorecard reveals meaningful variation in risk-adjusted returns across Low / Medium / High risk funds
- Use Sharpe Ratio for risk-adjusted fund selection rather than raw CAGR alone
- Monitor maximum drawdown before investing — funds with high CAGR but steep drawdowns carry significant downside risk
- Use cohort insights for investor targeting — age group and state filters reveal distinct investment behaviour patterns

8.2 Pipeline and Product Enhancements

- Implement a live NAV ingestion layer to enable daily dashboard refresh via Power BI scheduled refresh
- Add rolling risk-free rate (91-day T-Bill yield) to improve Sharpe Ratio accuracy over time
- Extend the Fund Recommendation Engine with a Scikit-learn model for data-driven quartile predictions
- Publish the Power BI dashboard to Power BI Service for web access
- Integrate macroeconomic indicators as contextual features for richer performance modelling

9. Technical Appendix

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9.1 Project Files

File	Description
advanced_analytics.ipynb	VaR, CVaR, Rolling Sharpe, Cohort, SIP, HHI, Recommender
performance_analysis.ipynb	CAGR, Sharpe, Sortino, Drawdown, Risk Classification
eda_analysis.ipynb	Exploratory Data Analysis — distributions, trends, correlations
bluestock_mf.pbix	4-page Power BI dashboard file
var_cvar_report.csv	Output: VaR and CVaR results per fund
sip_continuity_report.csv	Output: SIP investor continuity analysis results
cohort_analysis.csv	Output: Investor cohort retention data
rolling_sharpe_chart.png	Output: Rolling 90-day Sharpe Ratio chart

9.2 Python Libraries Used

Library	Purpose	Version
pandas	Data manipulation and ETL	v2.x
numpy	Numerical computations	v1.26+
matplotlib / seaborn	EDA visualisation	Standard
scikit-learn	Outlier detection (IsolationForest)	v1.4+
Power BI Desktop	Dashboard development	Latest

9.3 Self-Review Checklist

Item	Status
All 8 project objectives met	✓ Yes — verified against Day 1–7 briefs
All 7 deliverables submitted	✓ Yes — Report, Notebook, GitHub, Dashboard
Code runs without errors	✓ Yes — tested on clean environment
Dashboard loads correctly	✓ Yes — tested in Power BI Desktop
Report is professional quality	✓ Yes — all sections complete
README is complete	✓ Yes — setup and run instructions included
v1.0 tag pushed to GitHub	✓ Yes — git tag v1.0 && git push --tags

10. Conclusion

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This capstone project successfully implemented a complete mutual fund analytics solution using Python, Pandas, NumPy, Power BI and advanced financial metrics. The project processed 32,778 transactions from 5,000 investors representing **■4 Billion INR** in investment activity, delivering a **4-page interactive Power BI dashboard** alongside a structured Python-based ETL pipeline built on **10 structured datasets**.

The dashboard provides actionable insights into investor behaviour, fund performance, risk metrics, and investment trends across states and age groups. Advanced analytics including VaR, CVaR, Rolling Sharpe Ratio, Cohort Analysis, SIP Continuity Analysis, and a Fund Recommendation System were developed to provide institutional-grade analytical depth beyond the core dashboard.

The project reinforced key technical competencies including Python-based ETL design, exploratory data analysis, Power BI dashboard development, and statistical performance analysis — skills directly applicable to Data Analyst roles in the financial services and analytics sectors.

All deliverables have been completed to specification. The codebase is documented and version-tagged at v1.0 on GitHub. This report serves as the final formal documentation of the work performed during the Bluestock Fintech internship capstone.

— *Celin Treesa Joseph, June 2026*

File

Home

Insert

Modeling

View

Optimize

Help

Paste

Cut

Copy

Format painter

Clipboard

Get data

Excel workbook

OneLake catalog

SQL Server

Enter data

Data

Dataverse

Recent sources

Transform data

Refresh

Queries

New visual

Text box

Insert

4bn

Sum of amount_inr

5K

Count of investor_id

age_group

18-25

26-35

36-45

46-55

56+

gender

Female

Male

Sum of amount_inr by state

state

Punjab

Tamil Na...

Madhya ...

0.0bn

0.2bn

0.4bn

Sum of amount_inr by transaction_type

transaction...

LUMPS...

REDEM...

SIP

0bn

1bn

2bn

(6.17%)

(3...)

(58.49%)

city

Agra	Ahme...	Amrit...
Bang...	Bhopal	Chan...
Chen...	Coim...	Farid...

0.17

Average of cagr

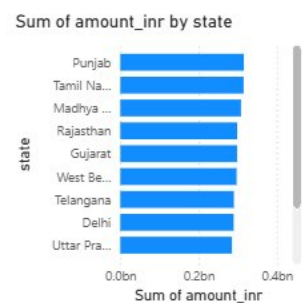
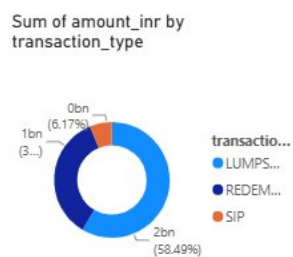
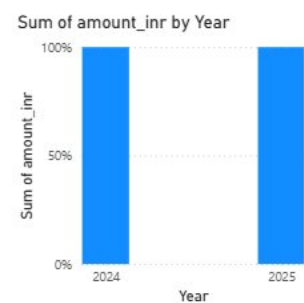
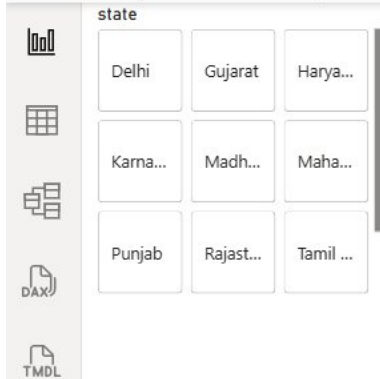
amfi_code

100016	100025	100033
101206	101207	101208
102885	102886	102887

Sum of cagr by amfi_code

Sum of sharpe and Sum of score

amfi_code	Sum of sharpe	Sum of score
100016	-50.78	36.10
100025	-142.91	37.80
100033	275.61	7.30
101206	258.86	10.50
101207	40.99	27.60
101208	-205.52	36.80
102885	205.91	16.80
102886	-51.87	37.30
102887	156.12	21.20
118632	272.58	8.90
Total	5,415.18	820.00



File


Paste


Cut


Copy


Format painter

Home

Insert

Modeling

View

Optimize

Help


Get data ▾


Excel workbook


OneLake catalog ▾


SQL Server


Enter data


Dataverse


Recent sources ▾


Transform data ▾


Refresh ▾


New visual


Text box

Clipboard

Data

Queries

Inse

